41 questions | **CLO** 4-A | **Midterm** 3/11/2026 · 6:15 pm

Q Exam-style questions: Sections 4-1 and 4-2

أسئلة بمعايير اختبارات جامعة السلطان قابوس — الاحتمالات 4-1 و 4-2 فقط

Covers only what sections 4-1 and 4-2 teach: sample spaces (lists, trees, dice tables), classical, empirical and subjective probability, the probability rules, complements, the phrases at least / at most / fewer than, mutually exclusive events, and addition rules 1, 2 and the three-event version. No conditional probability, multiplication rule or counting rules. Every question is tagged with its section.

طريقة الاستخدام

حل الجزء D أولًا (20 دقيقة) لتعرف مستواك، ثم الأجزاء A و B و C، ثم صحّح من ملف الإجابات. اكتب فضاء العينة والحدث **بصيغة المجموعات** دائمًا، وسمِّ القاعدة التي تستخدمها، فعليها درجات.

Part A · Multiple choice (16)

Give the letter and a one-line reason, as in a written exam.

A1 4-1

Two fair dice are rolled. The probability that the sum is a **prime number** is:

(a) 1/3 (b) 5/12
(c) 4/11 (d) 1/2

A2 4-1

Three fair coins are tossed. P(at least two heads) =

(a) 3/8 (b) 1/2
(c) 1/4 (d) 7/8

A3 4-1

Which value **cannot** be the probability of an event?

(a) 0 (b) 1
(c) −0.02 (d) 0.999

A4 4-1

The outcomes of an experiment have probabilities 0.20, 0.35, k and 0.15. The value of k is:

(a) 0.25 (b) 0.30
(c) 0.70 (d) cannot be found

A5 4-1

Of 1,200 SQU students surveyed, 360 said they use the bus, so P(bus) = 0.30. This is:

(a) classical probability
(b) empirical probability
(c) subjective probability
(d) a compound event

A6 4-1

A physician says there is a 25% chance a patient will need surgery, based on her experience. This is:

(a) classical
(b) empirical
(c) subjective
(d) impossible to classify

A7 4-1

If P(E) = 0.27, then P($\bar{E}$) is:

(a) 0.27
(b) 0.73
(c) −0.27
(d) 1.27

A8 4-1

Five students sit a quiz. The complement of "**at least 2** students pass" is:

(a) at most 2 pass
(b) fewer than 2 pass
(c) exactly 2 pass
(d) more than 2 fail

A9 4-1

A card is drawn from a standard 52-card deck. P(red face card) =

(a) 3/26
(b) 3/13
(c) 1/2
(d) 6/13

A10 4-1

When a single die is rolled, which event has probability 1?

(a) a number greater than 0
(b) an even number
(c) a 6
(d) a number greater than 6

A11 4-2

On one die, A = {2, 4, 6} and B = {1, 2, 3}. P(A or B) =

(a) 1
(b) 5/6
(c) 2/3
(d) 1/2

A12 4-2

A and B are mutually exclusive with P(A) = 0.45 and P(B) = 0.30. Then P(A ∪ B) and P(A ∩ B) are:

(a) 0.75 and 0.135 (b) 0.75 and 0

(c) 0.615 and 0 (d) 0.75 and 0.75

A13 4-2

P(A) = 0.6 and P(B) = 0.5. Which statement is **certainly true**?

(a) A and B are mutually exclusive (b) A and B cannot be mutually exclusive

(c) P(A ∪ B) = 1.1 (d) P(A ∩ B) = 0

A14 4-2

P(A ∪ B) = 0.7, P(A) = 0.4 and P(B) = 0.5. Then P(A ∩ B) =

(a) 0.2 (b) 0.1

(c) 0.3 (d) 0.9

A15 4-2

A card is drawn from a standard deck. P(queen or club) =

(a) 17/52 (b) 4/13

(c) 1/52 (d) 1/4

A16 4-2

P(A) = 0.5 and P(A ∩ B) = 0.2. The probability that A occurs but B does not is:

(a) 0.7 (b) 0.3

(c) 0.1 (d) 0.25

Part B · Short structured questions (14)

Typical 2–6 mark midterm items. Write S and E in set notation.

B1 4-1

[4 marks] A family has four children. (a) Write the sample space S and state n(S). (b) Find P(exactly two girls). (c) Find P(at least one boy). (d) Find P(the first and last child are of the same gender).

B2 4-1 4-2

[5 marks] Two fair dice are rolled. Find: (a) P(sum is 8) (b) P(sum is at most 4) (c) P(doubles or a sum of 8) (d) P(the two numbers differ by exactly 2) (e) P(sum is greater than 12).

B3 4-1

[4 marks] The number of visits 100 students made to the SQU clinic last semester:

VISITS	STUDENTS
0	12
1	30
2	25
3	18
4 or more	15

For a student chosen at random, find: (a) P(exactly 2 visits) (b) P(at most 1 visit) (c) P(at least 3 visits) (d) P(more than 1 but fewer than 4 visits).

B4 4-1 4-2

[5 marks] A number is chosen at random from S = {1, 2, ..., 15}. Let A = "a multiple of 3" and B = "an odd number". (a) Write A, B, $A \cap B$, $A \cup B$, $\bar{A}$ and $(A \cup B)^c$. (b) Find $P(A \cup B)$ using the addition rule and check it by counting. (c) Are A and B mutually exclusive? Explain.

B5 4-1 4-2

[6 marks] 200 SQU students were classified by college and transport:

	CAR	BUS	TOTAL
Science	30	40	70
Engineering	45	20	65
Arts	25	40	65
Total	100	100	200

A student is selected at random. Find: (a) P(Engineering) (b) P(Engineering and bus) (c) P(Engineering or bus) (d) P(Science or Arts) (e) P(neither Engineering nor car). (f) Are "Engineering" and "bus" mutually exclusive? Justify.

B6 4-2

[4 marks] P(A) = 0.35, P(B) = 0.45 and $P(A \cup B) = 0.62$. Find: (a) $P(A \cap B)$ (b) P(A but not B) (c) P(neither A nor B) (d) P(exactly one of A and B).

B7 4-2

[3 marks] In a phone shop, 32% of buyers choose Samsung, 41% Apple and 15% Huawei; each buyer buys one phone. Find: (a) P(Samsung or Apple) (b) P(another brand) (c) P(not Apple).

B8 4-2

[4 marks] Of 100 first-year students, 50 take Math, 40 Physics and 30 Chemistry; 15 take Math and Physics, 10 Math and Chemistry, 8 Physics and Chemistry, and 5 take all three. For a student chosen at random, find: (a) P(at least one of the three) (b) P(none) (c) P(Math only).

B9 4-1

[2 marks] Find the mistake. "Two coins are tossed. S = {two heads, one head, no heads}, so P(exactly one head) = 1/3."

B10 4-2

[2 marks] Find the mistake. "P(king or heart) = 4/52 + 13/52 = 17/52."

B11 4-1

[2 marks] An event E is three times as likely to occur as not to occur. Find P(E).

B12 4-2

[4 marks] In a company of 120 employees, 70 speak English, 45 speak French and 25 speak neither. An employee is chosen at random. Find: (a) P(speaks both) (b) P(speaks exactly one of the two) (c) P(speaks French only).

B13 4-1

[2 marks] Two dice are rolled. Find P(the product of the two numbers is even).

B14 4-2

[3 marks] Are the two events mutually exclusive? Explain. (a) A student is a STAT 2101 student; the student is a Science student. (b) Rolling one die: an odd number; a number greater than 5. (c) A card is a spade; the card is red.

Part C · Long questions (5)

Midterm-style, multi-part. Marks in brackets.

C1 4-1 4-2 **[12 marks]**

Blood types of 200 donors at the SQU Hospital blood bank:

	O	A	B	AB	TOTAL
Rh+	70	56	24	16	166
Rh−	14	10	6	4	34
Total	84	66	30	20	200

1. What type of probability is used here, and why?
2. P(Rh−)
3. P(type A or Rh−)
4. P(neither type O nor Rh+)
5. P(type A or type B); name the rule you used.
6. P(not AB)
7. Are "type O" and "Rh−" mutually exclusive?

C2 4-1 4-2 **[8 marks]**

A coin is tossed and then a die is rolled.

1. Draw a tree diagram and list the sample space. What is n(S)?
2. P(a head and an even number)
3. P(a head or an even number)
4. P(a tail and a number greater than 4)
5. Are "head" and "6" mutually exclusive?

C3 4-1 4-2 **[10 marks]**

Two fair dice, one red and one green, are rolled.

1. State n(S) and explain why the outcomes are equally likely.
2. P(the sum is odd)
3. P(the sum is odd or greater than 9)
4. P(the red die shows 6 or the sum is 7)
5. A player wins with a sum of 7 or 11. Find P(win) and P(lose).

C4 4-1 4-2 **[8 marks]**

Ages of 50 customers at a Muscat bank branch:

AGE	CUSTOMERS
18–25	9
26–33	14
34–41	12
42–49	10
50–57	5

1. P(a customer is 34–41)
2. P(a customer is at least 42)
3. P(a customer is younger than 34 or older than 49)
4. P(a customer is not in the 26–33 class)
5. Can you find P(a customer is younger than 30) from this table? Explain.

C5 4-2 **[10 marks]**

P(A) = 0.40, P(B) = 0.35 and P(A ∩ B) = 0.12.

1. Draw a Venn diagram showing the probability of each of the four regions.
2. P(A ∪ B)
3. P(A^c ∩ B^c)
4. P(A^c ∪ B)
5. P(exactly one of A, B)
6. Could P(A ∩ B) be 0.45 instead? Why not?

Part D · 20-minute mini-quiz (16 marks)

Time yourself, closed book, before opening the answers.

D1 4-1

[2 marks] Define *sample space* and *compound event*, with one example each.

D2 4-1

[3 marks] Three coins are tossed. Find P(exactly one tail), P(at most one tail) and P(no tails).

D3 4-2

[3 marks] P(A) = 0.25 and P(B) = 0.40. Find P(A ∪ B) if (a) A and B are mutually exclusive (b) P(A ∩ B) = 0.10.

D4 4-1 4-2

[4 marks] In a class, 18 of 30 students are female; 12 students wear glasses, 7 of them female. One student is chosen at random. Find P(female or wears glasses) and P(male without glasses).

D5 4-1

[2 marks] Two dice are rolled. Find P(the sum is at least 10).

D6 4-2

[2 marks] Explain, with an example, why mutually exclusive events cannot both occur, and write the addition rule they use.

Built to the STAT 2101 Fall 2026 course outline (CLO4-A: calculate probabilities using basic probability principles). Sections 4-1 Sample Spaces and Probability and 4-2 The Addition Rules only. Every numeric answer was computed by code from the question data.